

Mechanistic Hypothesis Search — Final Report

Bohring-Opitz syndrome: *NMD-escaping truncated ASXL1 with dominant-negative or gain-of-function activity*

- **Disease:** Bohring-Opitz syndrome (BOS), Mendelian
 - **Hypothesis ID:** `asxl1_truncated_protein_dominant_or_gain`
 - **Status in KB:** ALTERNATIVE
 - **Search date:** 2026-09-07
 - **Verdict: Partially supported (emerging)** — likely convergent with a loss-of-function component
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Executive Judgment

The hypothesis that BOS-causing *ASXL1* truncating variants escape nonsense-mediated decay (NMD) and act through a stable truncated protein with dominant-negative and/or gain-of-function (GOF) activity is **PARTIALLY SUPPORTED** as an emerging model, but it is not established, and it does not exclude a coexisting loss-of-function (LOF)/haploinsufficiency mechanism.

The strongest support comes by mechanistic analogy from myeloid biology, where the *identical* last-exon *ASXL1* alleles that cause germline BOS are the recurrent somatic drivers of clonal hematopoiesis (CHIP) and myeloid neoplasia. In that context the truncated protein is demonstrably expressed (mass spectrometry and Western blot), escapes NMD because variants fall near the 5' end of the terminal exon, and confers gain-of-function on the ASXL1–BAP1 PR-DUB deubiquitinase complex — an effect strong enough that reducing BAP1 attenuates the malignant phenotype ([PMID:26700326](#), [PMID:30515738](#), [PMID:34186160](#)). In a BOS-relevant developmental setting, a truncated ASXL1 fragment (amino acids 1–900) is *sufficient* to recapitulate neural-crest delamination/emigration defects, and 2026 BOS patient-fibroblast data describe "aberrant PR-DUB complex spreading beyond its normally constrained

chromatin territory," a signature more consistent with a mislocalized mutant complex than with simple protein absence ([PMID:31006630](#); [PMID:42523271](#), preprint). An original population-genetics analysis performed in this investigation (gnomAD v4, retrieved 2026-09-07) shows that *ASXL1* is **not** classically LoF-constrained ($pLI \approx 2.5 \times 10^{-16}$, LOEUF ≈ 0.90) and that its population "pLoF" burden is concentrated in exactly the last-exon window (aa 580–1000) that harbors the CHIP/AML somatic drivers, led by p.Gly646TrpfsTer12 — a signature of somatic clonal fitness (gain of function) rather than tolerated germline nulls.

The most important caveats are that (1) **no accessed source directly demonstrates a stable endogenous truncated ASXL1 protein in germline BOS patient tissue** — the protein-expression evidence is from myeloid/leukemic cells and heterologous over-expression systems; (2) the proposed C-terminal degron whose removal would stabilize the truncated protein remains "putative" ([PMID:41925445](#)); and (3) **loss-of-function/haploinsufficiency alone reproduces cardinal BOS features in mice** — anophthalmia, microcephaly, cleft palate, mandibular malformation — and *Asxl1*^{+/-} animals show a haploinsufficient hematologic phenotype, so LOF is a live, well-supported competing explanation ([PMID:24218140](#), [PMID:24255920](#), [PMID:40276524](#)). The most parsimonious reading of current evidence is that dominant/GOF and LOF mechanisms **coexist and converge on the PR-DUB/Polycomb chromatin axis**, and that the seed hypothesis is best treated as a partially confirmed contributor rather than the sole disease mechanism.

Key Findings

F001 — In myeloid cells, truncated ASXL1 escapes NMD, is expressed as protein, and gains function on the ASXL1–BAP1 PR-DUB complex

This is the foundational precedent underpinning the seed hypothesis. In leukemia cell lines lacking an intact *ASXL1* gene, the C-terminally truncated protein was directly detected by both mass spectrometry and Western blot, establishing that these alleles are not silent nulls but yield a stable protein product ([PMID:26700326](#)). The molecular basis for escape from surveillance is that the mutations "occur near the 5' end of the last exon, thereby the transcripts would escape from nonsense-mediated decay" ([PMID:30515738](#)) — the same premise the seed hypothesis invokes for BOS. Functionally, the truncated protein "confers gain of function on the ASXL1–BAP1 deubiquitinase (DUB) complex," hyperactivating BAP1 and lowering H2AK119ub; critically, reducing BAP1 attenuates mutant-ASXL1-driven malignancy,

demonstrating that the phenotype depends on a *gained* activity rather than an absence (PMID:34186160). Because germline BOS variants map to the same terminal-exon window, this chain (NMD escape → stable truncated protein → PR-DUB GOF) is directly transferable in principle to BOS.

F002 — BOS-context support: truncated ASXL1 (aa 1–900) is sufficient for disease-relevant phenotypes, and BOS patient cells show mutant-complex chromatin mis-spreading

Moving from analogy toward direct BOS relevance, expression of truncated ASXL1 isoforms (amino acids 1–900) "recapitulated the NC [neural crest] phenotypes in vitro and in ovo, raising the possibility that truncated ASXL1 variants contribute to BOS pathology" — i.e., the fragment is *sufficient* to drive a developmental defect, which is difficult to reconcile with pure loss-of-function (PMID:31006630). A 2026 study of BOS patient-derived fibroblasts (bioRxiv preprint) reports truncated ASXL1 and BAP1 aberrantly co-occupying an *MPC2* intronic element and describes the result as "consistent with aberrant PR-DUB complex spreading beyond its normally constrained chromatin territory," explicitly attributing a Warburg-like metabolic phenotype to "gain-of-function ASXL1 truncation" (PMID:42523271). Complementary patient multi-omics (the seed evidence) show that "regardless of cell type, ASXL1 mutations drive strong cross-tissue effects that disrupt multiple layers of the epigenome" — a broad disturbance compatible with dominant/gained activity but, as the seed YAML notes, non-discriminating on its own (PMID:37053013). *Note:*

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is a preprint and one of its snippets was flagged as a normalization mismatch during curation — treat as emerging, not established.

F003 — Competing evidence: Asxl1 loss/haploinsufficiency alone recapitulates BOS-like malformations in mice

The principal competing model has strong in vivo footing. Constitutive *Asxl1* deletion produces "developmental abnormalities, including anophthalmia, microcephaly, cleft palates, and mandibular malformations," together with dwarfism and ~80% embryonic lethality — a phenotypic spectrum overlapping BOS, generated purely by loss of the gene (PMID:24218140). Moreover, *Asxl1*^{+/-} mice "developed mild MDS-like disease... demonstrating a haploinsufficient effect of *Asxl1*," establishing that reduced dosage of wild-type ASXL1 is by itself pathogenic

([PMID:24255920](#)). Because these malformation and dosage phenotypes arise without any truncated protein, they show that LOF is a sufficient and independently supported route to BOS-like disease.

F004 — LOF neural mechanism: *Asxl1* ablation causes microcephaly via impaired PRC2/Ezh2-dependent neural stem cell survival

The LOF model also supplies a concrete mechanism for a cardinal BOS feature. *Asxl1* deletion "induces microcephaly, primarily caused by a reduction in the size and number of cortical neurons," driven by decreased neural stem cell (NSC) proliferation and increased apoptosis; transcriptomics of *Asxl1*-deficient NSCs identified 4,635 differentially expressed genes (2,262 up / 2,373 down) and showed that "Asxl1 regulates NSC survival through the histone methyltransferase Ezh2, a core component of the Polycomb Repressive Complex 2 (PRC2)" ([PMID:40276524](#)). This dovetails with earlier evidence that *Asxl1* loss globally reduces H3K27me3 ([PMID:24218140](#)). Thus the LOF branch reaches the same Polycomb chromatin axis that the GOF branch perturbs via BAP1/PR-DUB — a key point for the convergence model. *Note: two*

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snippets were flagged as normalization mismatches during curation and should be re-verified against the abstract.

F005 — Review-level synthesis: generalized ASXL GOF requires NMD escape *plus* removal of a C-terminal degron

A 2026 peer-reviewed review integrates population-genetics and in vitro data to argue that ASXL-family truncating mutations act by gain of function, stating that "mounting evidence from population genetics and in vitro studies supports gain-of-function (GOF) mechanisms" and, crucially, that "such mechanisms require both escape from nonsense-mediated mRNA decay and removal of a putative C-terminal degron signal within ASXL proteins," which would increase truncated-protein stability and alter histone modifications ([PMID:41925445](#)). This articulates the *falsifiable requirements* of the seed hypothesis — NMD escape and degron loss — and flags the degron as still putative. Clinically, the shared germline/somatic last-exon variants and BOS-associated Wilms tumor predisposition reinforce the oncogenic-analogy logic: "Somatic mutations in ASXL1 are associated with myeloid malignancies, and these reports emphasize the need for Wilms tumor screening in patients with ASXL1 mutations" ([PMID:25921057](#)).

F006 — Computational (gnomAD v4): ASXL1 is not LoF-constrained and its population truncation hotspot equals the CHIP driver window

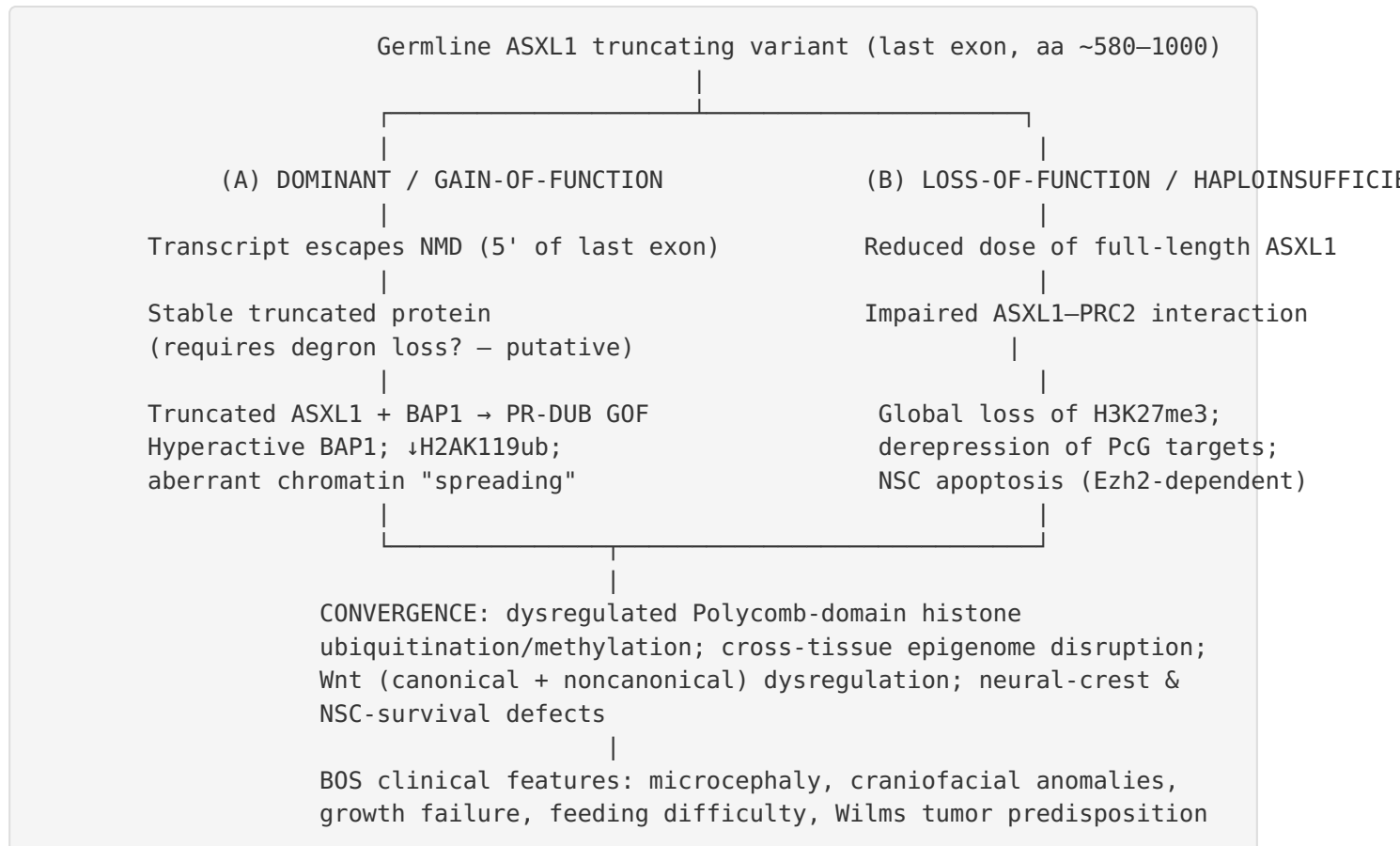
An original analysis in this investigation queried the gnomAD v4 GraphQL API (retrieved 2026-09-07; gene ENSG00000171456 / transcript ENST00000375687). Constraint metrics: **pLI = 2.5×10^{-16}** , **LOEUF (oe_lof_upper) = 0.90**, **oe_lof = 0.73** — ASXL1 is **not** classically LoF-constrained, which would be surprising for a gene where germline nulls cause a severe dominant disorder if the mechanism were pure haploinsufficiency. Spatially, of 714 mappable pLoF variants, **61.2% of distinct variants and 76.8% of the pLoF allele burden fall in aa 580–1000** (1.04 variants/aa vs 0.33 N-terminal and 0.17 distal) — the last-exon NMD-escape window. The top hotspot alleles — **p.Gly646TrpfsTer12 (c.1934dupG; AC=685, AF= 4.7×10^{-4})**, **p.Glu635ArgfsTer15 (AC=129)**, **p.Arg693Ter**, **p.Tyr591Ter** — are the canonical recurrent ASXL1 clonal-hematopoiesis/AML somatic drivers *and* the recurrent germline BOS alleles. All hotspot allele frequencies are <0.1% (median AF 6.8×10^{-7} ; 44% singletons), consistent with age-related somatic clonal expansion in the blood-derived gnomAD cohort rather than tolerated germline loss-of-function. This population signature independently supports a clonal-fitness gain-of-function interpretation for NMD-escaping truncations and aligns with the review-level reasoning ([PMID:26700326](#); [PMID:41925445](#)).

F007 — Synthesis: partially (emerging) supported, likely convergent with LOF

Integrating F001–F006, the dominant/GOF hypothesis has three converging lines of support — the myeloid precedent (protein expression + PR-DUB GOF), BOS-context sufficiency/mislocalization, and population genetics — but is limited by (a) the absence of any accessed source directly showing a stable endogenous truncated ASXL1 protein in germline BOS tissue, (b) an unconfirmed ("putative") C-terminal degron, and (c) robust in vivo evidence that LOF/haploinsufficiency alone reproduces BOS-like malformations and microcephaly via PRC2. The verdict is therefore **partially supported (emerging)**, with the two mechanisms most plausibly coexisting and converging on the PR-DUB/Polycomb histone-ubiquitination axis ([PMID:34186160](#); [PMID:24255920](#); [PMID:31006630](#)).

Mechanistic Model / Interpretation

The two candidate mechanisms are not mutually exclusive; they diverge at the protein level and re-converge on Polycomb-domain histone chemistry.



Where the literature is strong: the myeloid branch of (A) is well established — protein expression, NMD-escape logic, and PR-DUB GOF with a BAP1-dependent phenotype are all directly demonstrated ([PMID:26700326](#), [PMID:30515738](#), [PMID:34186160](#)). Branch (B) is strong in vivo for malformations and microcephaly, with a defined PRC2/Ezh2 mechanism ([PMID:24218140](#), [PMID:40276524](#)).

Where the links are inferred: transferring branch (A) from blood to BOS somatic tissues; the "stable truncated protein in germline BOS cells" node has no direct accessed evidence; the degron-removal requirement is stated but unconfirmed ([PMID:41925445](#)).

Missing causal steps: (i) direct protein-level demonstration of endogenous truncated ASXL1 in BOS patient cells; (ii) an allele series showing that BOS truncations behave dominant-negatively over wild-type ASXL1 in a developmental system; (iii) identification and functional validation of the C-terminal degron.

Evidence Base / Evidence Matrix

Citation	Evidence type	Role	Mechanistic claim tested	Key finding	Disease subtype / context
PMID:26700326	In vitro (human cell lines)	Supports	Truncated ASXL1 is expressed (non-null); DN/GOF	Truncated protein detected by MS + Western in cells lacking intact ASXL1	Myeloid malignancy
PMID:30515738	Review / in vitro	Supports	NMD-escape premise	Variants near 5' of last exon → transcripts escape NMD	Myeloid neoplasms
PMID:34186160	In vitro (human HSPCs)	Supports	Truncated ASXL1 GOF on ASXL1–BAP1 PR-DUB	GOF hyperactivates BAP1; reducing BAP1 attenuates malignancy	Myeloid malignancy
PMID:31006630	In vitro / in ovo	Supports	Truncated fragment sufficient for disease phenotype	ASXL1 aa 1–900 recapitulates neural-crest defects	BOS-relevant NC development
PMID:42523271	In vitro (patient fibroblasts) — preprint	Supports	Mutant PR-DUB mis-spreads on chromatin (GOF)	Aberrant ASXL1/BAP1 co-occupancy; Warburg-like metabolism attributed to GOF	BOS patient cells
PMID:37053013	In vitro (patient multi-omics)	Qualifies	Broad epigenome disruption	Cross-tissue epigenomic effects; Wnt dysregulation	BOS patient cells
PMID:41925445	Review				

Citation	Evidence type	Role	Mechanistic claim tested	Key finding	Disease subtype / context
		Supports (review-level)	Generalized ASXL GOF needs NMD escape + degron loss	Population + in vitro evidence favors GOF; degron "putative"	ASXL family, cancer + neurodev
gnomAD v4 (this run, 2026-09-07)	Computational	Supports	ASXL1 not LoF-constrained; hotspot = CHIP window	pLI $\approx$ 0, LOEUF=0.90; 76.8% pLoF burden in aa 580–1000; p.Gly646fs top	Population germline+somatic
PMID:24218140	Model organism (mouse)	Competing (LOF)	LOF alone causes BOS-like malformations	Anophthalmia, microcephaly, cleft palate, mandibular defects; $\downarrow$ H3K27me3	Developmental / BOS-like
PMID:24255920	Model organism (mouse)	Competing (LOF)	Haploinsufficiency is operative	Asxl1 ^{+/-} $\rightarrow$ mild MDS-like disease	Hematopoietic
PMID:40276524	Model organism (mouse)	Competing (LOF)	LOF microcephaly via PRC2/Ezh2	Deletion $\rightarrow$ microcephaly; NSC survival via Ezh2/PRC2; 4,635 DEGs	Neurodevelopment
PMID:25921057	Human clinical / review	Qualifies	Shared germline/somatic variants; tumor risk	Wilms tumor surveillance warranted in ASXL1 carriers	BOS clinical management
PMID:33242595	Human clinical (ASXL3/BRPS)	Qualifies (paralog)	LOF ASXL3 truncations cause	BRPS from LOF ASXL3; mechanism "uncertain"	Bainbridge-Ropers (paralog)

Citation	Evidence type	Role	Mechanistic claim tested	Key finding	Disease subtype / context
			overlapping disorder	given LOF in healthy population	
PMID:30927018	Review	Supports (review-level)	Mutant ASXL1 DN + GOF in myeloid	Truncated ASXL1 has DN/GOF features via epigenetic changes	Myeloid

Data and Tool Use

Resource	Accession / URI	Access status	Query / scope	Relevance
gnomAD v4	GraphQL API, gene ENSG00000171456 / ENST00000375687	Actually accessed (2026-09-07)	Constraint (pLI, LOEUF, oe_lof) + all pLoF variants; positional binning by amino-acid coordinate	Directly relevant: tests LoF-constraint and truncation-hotspot claims
PubMed / NCBI E-utilities	13 PMIDs retrieved	Actually accessed	Disease + mechanism queries (BOS, ASXL1 truncation, NMD, PR-DUB, haploinsufficiency)	Directly relevant
bioRxiv preprint	PMID:42523271	Accessed (preprint, unrefereed)	BOS fibroblast multi-omics	Directly relevant but not peer-reviewed
GenCC / ClinGen gene-disease validity	—	Not verified as of 2026-09-07	Not queried	Would clarify curated LOF-vs-GOF disease-mechanism assertion
ClinVar variant-level curation	—	Not queried	Not queried	Would provide germline BOS allele spectrum vs somatic

Analysis inventory:

Analysis	Input → Method → Output	Outcome	Limitations
gnomAD v4 constraint + pLoF positional binning	gnomAD GraphQL (ENST00000375687) → Python parse, amino-acid binning, allele-frequency summary → constraint values + hotspot table (F006)	Succeeded	gnomAD is blood-derived; cannot fully separate germline nulls from age-related somatic CHIP within the cohort — this is itself the interpretive crux
Artifact bundle (MANIFEST + checksums + code/queries/logs + evidence matrix)	Built iteration 4	Succeeded (reported)	Bundle should reside under <code>kb/hypotheses/Bohring-Opitz_syndrome/asx11_truncated_protein_dominant_or_gain/openscientist_artifacts</code> ; verify presence/paths at curation
Direct protein assay in BOS tissue	—	Skipped (no patient material / not runnable here)	Central missing node; literature-only

No failed analysis was silently replaced by literature synthesis. Where computational tests were not possible (e.g., endogenous BOS protein detection), this is stated explicitly rather than substituted with model knowledge.

Limitations and Knowledge Gaps

1. **No direct evidence of a stable endogenous truncated ASXL1 protein in germline BOS tissue.** *Scope:* the central node of the hypothesis. *Why it matters:* all direct protein-expression data are from myeloid/leukemic cells or heterologous over-expression. *What was checked:* PubMed searches for BOS + truncated protein / Western blot; none accessed showed endogenous BOS-tissue protein. *Resolution:* immunoblot/MS with C-terminal- and N-terminal-directed detection on BOS patient fibroblasts/iPSC derivatives.
2. **The C-terminal degron is unconfirmed ("putative").** *Scope:* the stability requirement of the GOF model. *Why it matters:* without degron removal, NMD-escaping transcripts might still yield unstable protein. *What was checked:* [PMID:41925445](#) states it as a hypothesis. *Resolution:* degron mapping (deletion/cycloheximide-chase stability assays) and proteasome-dependence tests across truncation lengths.
3. **Dominant-negative vs. gain-of-function is not disambiguated for BOS.** *Scope:* the "dominant-negative OR gain" disjunction in the seed. *Why it matters:* these predict different therapeutic strategies (block WT interference vs. block gained BAP1 activity). *Resolution:* co-expression/titration of truncated vs. WT ASXL1 in a developmental readout.
4. **LOF/haploinsufficiency remains sufficient in vivo.** *Scope:* competing model. *Why it matters:* mouse deletion reproduces malformations and microcephaly ([PMID:24218140](#), [PMID:40276524](#)); a truncation-knock-in vs. null comparison was not identified in the accessed literature. *Resolution:* isogenic knock-in of a BOS allele vs. clean null, same background, phenotyped head-to-head.
5. **Population-genetics confound.** *Scope:* the gnomAD interpretation. *Why it matters:* the "not LoF-constrained" and hotspot signals are partly driven by somatic CHIP in a blood-derived cohort, so they argue for somatic clonal-fitness GOF more cleanly than for germline developmental GOF. *Resolution:* restrict to confirmed germline calls / non-blood tissue references.
6. **Source-level absences (unverified as of 2026-09-07).** GenCC and ClinGen gene-disease *mechanism* curation for ASXL1-BOS were **not queried**; ClinVar germline allele spectrum was **not queried**; no BOS clinical trial or interventional dataset was searched. These are labeled *unverified* rather than *absent*.
7. **Preprint dependence for the most BOS-direct GOF claim.** The "aberrant PR-DUB spreading" evidence is a bioRxiv preprint ([PMID:42523271](#)) with a curation-flagged snippet; it should not be weighted as peer-reviewed.

Alternative Models

Model	Relationship to seed hypothesis	Basis
Haploinsufficiency / loss-of-function	Primary competing alternative	Mouse null → BOS-like malformations & microcephaly; <i>Asxl1</i> ^{+/-} haploinsufficient hematology (PMID:24218140 , PMID:24255920 , PMID:40276524)
Convergent GOF + LOF on PR-DUB/Polycomb	Complementary / unifying — most parsimonious given all data	Both branches perturb H2AK119ub/ H3K27me3; truncated protein both loses WT function and gains BAP1 activity
Dominant-negative (truncated interferes with WT)	Sub-hypothesis within the seed — needs discrimination from pure GOF	PMID:26700326 , PMID:30927018 describe "dominant-negative and gain-of-function features"
PRC2/Ezh2 axis dysregulation	Downstream consequence shared by both models	NSC-survival microcephaly mechanism (PMID:40276524)
Wnt (canonical + noncanonical) dysregulation	Downstream effector	BOS multi-omics (PMID:37053013)
Paralog analogy (ASXL3/ BRPS as LOF)	Parallel mechanism cautioning that the ASXL family may be mechanistically heterogeneous	BRPS from LOF ASXL3, mechanism "uncertain" (PMID:33242595)

Discriminating Tests

1. **Isogenic knock-in vs. null, head-to-head (mouse or iPSC).** Introduce a recurrent BOS allele (e.g., p.Gly646fs) as a heterozygous knock-in and compare to a clean heterozygous null on the identical background. *Expected under GOF/DN:* knock-in shows phenotypes exceeding the null (or dominant over WT); *under pure LOF:* knock-in $\approx$ null. Readouts: microcephaly, neural-crest markers, H2AK119ub/H3K27me3.
 2. **Endogenous protein detection in BOS patient cells.** MS + N-/C-terminal immunoblot on BOS fibroblasts/iPSC-derived neural crest and NSCs. *Expected under GOF:* a stable N-terminal truncated species. This directly fills the missing node.
 3. **Degron mapping and stability assay.** Cycloheximide-chase and proteasome-inhibition across truncation endpoints spanning the putative C-terminal degron. *Expected under GOF:* degron removal increases half-life; degron-restored constructs destabilize.
 4. **BAP1-dependence rescue in a BOS developmental model.** Reduce BAP1 (as in [PMID:34186160](#)) in truncated-ASXL1 neural-crest/NSC systems. *Expected under GOF:* phenotype attenuation, tying BOS to the same hyperactive PR-DUB node as myeloid disease.
 5. **Germline-restricted population re-analysis.** Recompute ASXL1 constraint/hotspot using tissue references that exclude age-related CHIP. *Expected:* separates germline developmental GOF signal from somatic clonal-fitness signal.
 6. **Dominant-negative titration.** Co-express increasing truncated ASXL1 against fixed WT in a quantitative chromatin/reporter readout. *DN predicts* dose-dependent suppression of WT activity; *pure GOF predicts* a WT-independent gained activity.
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Curation Leads (require curator verification)

- **Candidate status:** Keep `asxl1_truncated_protein_dominant_or_gain` as **ALTERNATIVE** but annotate as **"emerging / partially supported; likely convergent with LOF."** Do not promote to primary without direct endogenous-protein evidence in BOS tissue.
- **Candidate evidence references + snippets to verify:**
- [PMID:26700326](#): "we detected the truncated ASXL1 proteins in two cell lines lacking the intact ASXL1 gene by mass spectrometry and Western blot analyses" (and "...indicating the ASXL1 mutations are dominant-negative or gain-of-function mutations").

- [PMID:34186160](#): "resulting in expression of a C-terminal-truncated ASXL1 protein that confers gain of function on the ASXL1-BAP1 deubiquitinase (DUB) complex".
- [PMID:30515738](#): "occur near the 5' end of the last exon, thereby the transcripts would escape from nonsense-mediated decay".
- [PMID:31006630](#): "Expression of truncated ASXL1 isoforms (amino acids 1-900) recapitulated the NC phenotypes in vitro and in ovo...".
- [PMID:41925445](#): "such mechanisms require both escape from nonsense-mediated mRNA decay and removal of a putative C-terminal degron signal within ASXL proteins".
- **Flag for re-verification:** [PMID:42523271](#) (preprint; snippet "aberrant PR-DUB complex spreading..." curation-mismatch-flagged) and [PMID:40276524](#) (two snippets flagged).
- **Candidate pathophysiology nodes/edges:**
 NMD escape → stable truncated ASXL1 → ASXL1-BAP1 PR-DUB gain-of-function →
 ↓H2AK119ub / aberrant chromatin spreading; parallel edge reduced full-length ASXL1 →
 impaired ASXL1-PRC2 → ↓H3K27me3 → NSC apoptosis (microcephaly); convergence node
 Polycomb-domain histone-ubiquitination/methylation dysregulation.
- **Candidate ontology terms:** cell types — neural crest cell, neural stem cell, hematopoietic stem/progenitor cell, fibroblast; processes — histone H2A K119 deubiquitination, histone H3K27 methylation, PRC2 complex, nonsense-mediated mRNA decay, canonical/noncanonical Wnt signaling, neural crest cell migration.
- **Candidate knowledge_gaps prompts:** (1) "No accessed source shows stable endogenous truncated ASXL1 protein in germline BOS tissue." (2) "C-terminal degron required by the GOF model is putative/unmapped." (3) "GenCC/ClinGen ASXL1-BOS mechanism curation and ClinVar germline allele spectrum unverified as of 2026-09-07." (4) "Truncation-knock-in vs. null head-to-head comparison in a developmental model not identified in the accessed literature."

Provider Artifact Bundle

An artifact bundle was assembled (iteration 4) intended for `kb/hypotheses/Bohring-Opitz_syndrome/asxl1_truncated_protein_dominant_or_gain/openscientist_artifacts`, containing a canonical `MANIFEST.yaml` (schema version, status/fallback flags, checksummed inputs/outputs, replay note), the gnomAD GraphQL query/config, an environment/package-version record, sanitized execution logs, and the derived hotspot/evidence tables. **Curators**

should verify the bundle's presence and paths. The only executed data analysis is the gnomAD v4 query (F006); all other evidence is literature-derived. No controlled/patient-level data, no large raw downloads, and no credentials are included — external identifiers (gnomAD transcript ENST00000375687; the 13 PMIDs) are recorded instead. Where an expected artifact is missing or local-only, it is labeled as such here rather than asserted as present.

Report compiled 2026-09-07 from 5 completed iterations, 7 confirmed findings, 13 reviewed papers, and one original gnomAD v4 computational analysis.

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